Linux commands

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Scenario 1 – Company File Organization

```
root@f0ee46022980: /compai
ubuntu:latest   d1e2e92c075e        119MB         31.7MB

C:\Users\fasee>docker pull kalilinux/kali-rolling
Using default tag: latest
latest: Pulling from kalilinux/kali-rolling
194d70789344: Pull complete
Digest: sha256:287cd5cfa409e258e9ec3661db4dff0bfbb45fc95734e82d3b270a0f74962
9ca
Status: Downloaded newer image for kalilinux/kali-rolling:latest
docker.io/kalilinux/kali-rolling:latest

C:\Users\fasee>docker images

                        i Info →   U  In Use
IMAGE                           ID             DISK USAGE   CONTENT SIZE   E
XTRA
kalilinux/kali-rolling:latest   287cd5cfa409        187MB         52.2MB
ubuntu:latest                   d1e2e92c075e        119MB         31.7MB

C:\Users\fasee>docker run -it kalilinux/kali-rolling:latest
┌──(root㉿f0ee46022980)-[/]
└─# mkdir company

┌──(root㉿f0ee46022980)-[/]
└─# cd company

┌──(root㉿f0ee46022980)-[/company]
└─# mkdir hr

┌──(root㉿f0ee46022980)-[/company]
└─# mkdir finance

┌──(root㉿f0ee46022980)-[/company]
└─# mkdir it

┌──(root㉿f0ee46022980)-[/company]
└─# ls
finance  hr  it

┌──(root㉿f0ee46022980)-[/company]
└─#
```

```
┌──(root㉿f0ee46022980)-[/company]
└─# cd hr

┌──(root㉿f0ee46022980)-[/company/hr]
└─# touch employees.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# ls
employees.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# touch salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# ls
employees.txt  salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

```
┌──(root㉿f0ee46022980)-[/company/hr]
└─# apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB
]
Get:4 http://kali.download/kali kali-rolling/non-free-firmware amd64 Package
s [14.3 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Fetched 21.3 MB in 13s (1675 kB/s)
All packages are up to date.

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

```
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the
 Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_
64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-
gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib
/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) a
t /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 3.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package libncursesw6:amd64.
(Reading database ... 5144 files and directories currently installed.)
Preparing to unpack .../libncursesw6_6.6+20251231-1_amd64.deb ...
Unpacking libncursesw6:amd64 (6.6+20251231-1) ...
Selecting previously unselected package nano.
Preparing to unpack .../nano_8.7.1-1_amd64.deb ...
Unpacking nano (8.7.1-1) ...
Selecting previously unselected package libgpm2:amd64.
Preparing to unpack .../libgpm2_1.20.7-12+b1_amd64.deb ...
Unpacking libgpm2:amd64 (1.20.7-12+b1) ...
Setting up libgpm2:amd64 (1.20.7-12+b1) ...
Setting up libncursesw6:amd64 (6.6+20251231-1) ...
Setting up nano (8.7.1-1) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in
auto mode
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto
 mode
Processing triggers for libc-bin (2.42-13) ...

┌──(root㉿f0ee46022980)-[/company/hr]
└─# nano --version
 GNU nano, version 8.7.1
 (C) 2026 the Free Software Foundation and various contributors
 Compiled options: --disable-libmagic --enable-utf8

┌──(root㉿f0ee46022980)-[/company/hr]
└─# 
```

```
root@f0ee46022980: /compai

  GNU nano 8.7.1                    employees.txt *
Ali
Sara
Ahmed

                              [ Read 0 lines ]
^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
┌──(root㉿f0ee46022980)-[/company/hr]
└─# nano employees.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# cat employees.txt
Ali
Sara
Ahmed

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

Scenario 2 – Searching Important Data

```
┌──(root㉿f0ee46022980)-[/company/hr]
└─# cat employees.txt
Ali
Sara
Ahmed

┌──(root㉿f0ee46022980)-[/company/hr]
└─# grep Ahmed employees.txt
Ahmed

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

```
┌──(root㉿f0ee46022980)-[/company/hr]
└─# wc -l employees.txt
3 employees.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

```
┌──(root㉿f0ee46022980)-[/company/hr]
└─# cat employees.txt
Ali
Sara
Ahmed

┌──(root㉿f0ee46022980)-[/company/hr]
└─# grep Ahmed employees.txt
Ahmed

┌──(root㉿f0ee46022980)-[/company/hr]
└─# wc -l employees.txt
3 employees.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# head -n 1 employees.txt
Ali

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

Scenario 3 – Backup Operation

```
┌──(root㉿f0ee46022980)-[/company]
└─# mkdir backup

┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup  finance  hr  it

┌──(root㉿f0ee46022980)-[/company]
└─# cd hr

┌──(root㉿f0ee46022980)-[/company/hr]
└─# ls
employees.txt  salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# cp employees.txt ../backup/

┌──(root㉿f0ee46022980)-[/company/hr]
└─# cd ..

┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup  finance  hr  it

┌──(root㉿f0ee46022980)-[/company]
└─# cd backup

┌──(root㉿f0ee46022980)-[/company/backup]
└─# ls
employees.txt
```

```
┌──(root㉿f0ee46022980)-[/company/backup]
└─# cat employees.txt
Ali
Sara
Ahmed

┌──(root㉿f0ee46022980)-[/company/backup]
└─#
```

Scenario 4 – File Permissions

```
┌──(root㉿f0ee46022980)-[/company/hr]
└─# nano salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# cat salaries.txt
Confidential Salary Data

┌──(root㉿f0ee46022980)-[/company/hr]
└─# ls -la
total 16
drwxr-xr-x 2 root root 4096 Mar 25 19:40 .
drwxr-xr-x 6 root root 4096 Mar 25 18:53 ..
-rw-r--r-- 1 root root   15 Mar 25 18:50 employees.txt
-rw-r--r-- 1 root root   25 Mar 25 19:40 salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# chmod 600 salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─# ls -la
total 16
drwxr-xr-x 2 root root 4096 Mar 25 19:40 .
drwxr-xr-x 6 root root 4096 Mar 25 18:53 ..
-rw-r--r-- 1 root root   15 Mar 25 18:50 employees.txt
-rw------- 1 root root   25 Mar 25 19:40 salaries.txt

┌──(root㉿f0ee46022980)-[/company/hr]
└─#
```

Scenario 5 – System Log Analysis

```
┌──(root㉿f0ee46022980)-[/company]
└─# touch server.log

┌──(root㉿f0ee46022980)-[/company]
└─# nano server.log

┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup  finance  hr  it  server.log

┌──(root㉿f0ee46022980)-[/company]
└─# cat server.log
login success
login failed
disk error
login success
network error

┌──(root㉿f0ee46022980)-[/company]
└─# |
```

```
┌──(root㉿f0ee46022980)-[/company]
└─# cat server.log
login success
login failed
disk error
login success
network error

┌──(root㉿f0ee46022980)-[/company]
└─# grep -c "login" server.log
3

┌──(root㉿f0ee46022980)-[/company]
└─# grep error server.log
disk error
network error

┌──(root㉿f0ee46022980)-[/company]
└─#
```

Scenario 6 – Power User Challenge

```
┌──(root㉿f0ee46022980)-[/company]
└─# mkdir tests

┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup  finance  hr  it  server.log  tests

┌──(root㉿f0ee46022980)-[/company]
└─# touch test{1..10}.txt

┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup   it          test10.txt  test4.txt  test7.txt  tests
finance  server.log  test2.txt   test5.txt  test8.txt
hr       test1.txt   test3.txt   test6.txt  test9.txt

┌──(root㉿f0ee46022980)-[/company]
└─#
```

```
┌──(root㉿f0ee46022980)-[/company]
└─# mv test*.txt tests/

┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup  finance  hr  it  server.log  tests

┌──(root㉿f0ee46022980)-[/company]
└─# cd tests

┌──(root㉿f0ee46022980)-[/company/tests]
└─# ls
test1.txt   test2.txt  test4.txt  test6.txt  test8.txt
test10.txt  test3.txt  test5.txt  test7.txt  test9.txt

┌──(root㉿f0ee46022980)-[/company/tests]
└─# |
```

```
┌──(root㉿f0ee46022980)-[/company]
└─# ls
backup  finance  hr  it  server.log  tests

┌──(root㉿f0ee46022980)-[/company]
└─# tree
.
├── backup
│   └── employees.txt
├── finance
├── hr
│   ├── employees.txt
│   └── salaries.txt
├── it
├── server.log
└── tests
    ├── test1.txt
    ├── test10.txt
    ├── test2.txt
    ├── test3.txt
    ├── test4.txt
    ├── test5.txt
    ├── test6.txt
    ├── test7.txt
    ├── test8.txt
    └── test9.txt

6 directories, 14 files

┌──(root㉿f0ee46022980)-[/company]
```